

EXPERIMENTAL AND QUASI-EXPERIMENTAL DESIGNS, CHAPTER 1

Experiments and Generalized Causal Inference

A Plain-Language Guide

What makes something a cause, what counts as a real experiment, and how far a single study's findings can be trusted to generalize.

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Based on Shadish, Cook, and Campbell, “Experiments and Generalized Causal Inference,” in *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*

What This Chapter Covers

This chapter lays the conceptual groundwork for everything that follows: what causation actually means, the specialized vocabulary of experiments, how findings from one study generalize to others, and where experimentation fits within science as a whole.

Four questions guide the chapter: What is the nature of the causation that experiments test? What specialized terms describe different kinds of studies? How do we generalize a causal connection beyond the single study that found it? And how does experimentation fit into the larger philosophy and sociology of science?

FOUNDATIONS, 2 OF 3

The History of Experimentation

Deliberate experimenting is as old as humanity — people have always tried new recipes or new ways of starting a fire. But modern scientific experimentation, which emerged in the 16th and 17th centuries, changed the practice in three ways.

Observation began to be used to correct errors in theory rather than just support it. Researchers started deliberately manipulating the world instead of only passively observing it. And scientists began working to control extraneous influences that might bias what they observed — from clearer telescope lenses to sterilized test tubes.

FOUNDATIONS, 3 OF 3

What Makes Something an Experiment?

As new fields like public health and education began experimenting, laboratory-style controls worked poorly, so researchers developed new tools for the job: random assignment and nonrandomized control groups.

The feature common to all experiments, then and now, is deliberately varying something to discover what happens to something else later. This book focuses on field experimentation in the social sciences — a specialized extension of something humans have always done.

Defining Cause, Effect, and Causal Relationship

People intuitively recognize causal relationships in daily life — a low test grade can be both a cause and an effect, for instance, of not studying. But a precise definition of cause and effect has eluded philosophers for centuries.

This chapter doesn't claim to offer the one true definition. Instead, it works with ideas about cause, effect, and causal relationship that are specific enough to clarify how experiments actually work.

What Is a Cause?

Consider a forest fire. A match, a lightning strike, or a smoldering campfire can each start one — but none is necessary, since a fire can start other ways, and none is sufficient alone.

A lighted match needs dry leaves, oxygen, and dry weather to start a fire. Philosopher John Mackie called this an inus condition — an insufficient but non-redundant part of a condition that is itself unnecessary but sufficient. Most causes work this way: they only produce their effect as part of a larger, often only partly understood, constellation of conditions. Most causal relationships are therefore probabilistic, not deterministic: a cause changes the likelihood of an outcome, and whether it does so depends on context. That context-dependence is why causal generalization matters.

The Endostatin Example

In the late 1990s, a Boston research team reported that a new drug called Endostatin shrank tumors by limiting their blood supply. Other labs using drug shipped from the original team couldn't replicate the effect.

Scientists eventually succeeded only after traveling to the original lab to learn exactly how to manufacture, transport, handle, and inject the drug. One observer called these hidden contingencies the “in-our-hands” phenomenon: Endostatin's effectiveness depended on a larger set of conditions that even the original investigators didn't fully understand — a real-world inus condition.

What Is an Effect?

An effect is best understood through a counterfactual: something contrary to fact. In an experiment, we observe what did happen to people who received a treatment; the counterfactual is what would have happened to those same people if, at the same time, they had not.

An effect is the difference between what did happen and what would have happened. But we can never actually observe a counterfactual — the same person cannot simultaneously receive and not receive a treatment. So the central task of all cause-probing research is to construct a reasonable approximation of that physically impossible comparison. This counterfactual logic is also the foundation of what statisticians call Rubin's Causal Model.

Approximating the Counterfactual

Consider phenylketonuria (PKU), a genetic disease that causes mental retardation unless treated early with a restricted diet. We can't observe what would have happened to the same infants without the diet — so researchers build approximations instead.

We might compare treated infants with similar untreated infants — but then the two groups are different people. Or we might compare the same infants before and after starting the diet — but then time has passed and other things have changed. Neither is a true counterfactual. Creating the best available approximation, and understanding exactly how it falls short, are two central tasks of experimental design.

When Is a Relationship Causal?

Philosopher John Stuart Mill proposed a classic three-part test for a causal relationship.

The cause must precede the effect. The cause must be related to the effect. And there must be no plausible alternative explanation for the effect other than the cause. These three conditions mirror exactly what happens in an experiment: researchers manipulate a presumed cause, observe whether the effect varies accordingly, and use various methods to rule out other explanations.

Correlation, Confounds, and the Third Variable

Correlation does not prove causation — a familiar maxim, but a real methodological problem. We may not know which of two correlated variables came first, or whether a third variable explains both.

Income and education are correlated: does income buy education, or does education lead to higher income? Both could be true. Or a confound — a third variable like intelligence — might cause both high education and high income without either causing the other. Identifying the confounds that can operate in a given research area is a central task in the study of experiments.

Manipulable and Nonmanipulable Causes

It makes sense to ask what happens if welfare recipients are required to work. It makes no sense to ask what happens if we change an adult into a three-year-old. Experiments can only manipulate causes like the first kind.

Nonmanipulable events, like a supernova, or attributes, like a person's age or biological sex, cannot be varied to see what happens — so most scientists agree it's much harder to discover their effects with confidence. Even a genetic defect like the one behind PKU, though not directly manipulable, can be studied indirectly by manipulating something downstream of it, like diet.

Studying the Nonmanipulable

Manipulable causes have two real advantages over nonmanipulable ones: they offer a direct action to solve a problem, and they allow a higher-quality counterfactual through methods like random assignment.

Still, nonmanipulable causes are worth studying however possible, since they often point the way toward manipulable ones — PKU research had to identify the nonmanipulable genetic defect before scientists could find the manipulable diet that treats it. Analogue experiments (inducing skin pigmentation changes to study race) and natural experiments (like the Vietnam-era draft lottery) offer other ways in.

What Experiments Cannot Do

Experimenting on manipulable causes makes discovering their effects easier, but experiments are far from a perfect tool. They tell us nothing about how or why an effect occurs, and they can't answer many real-world questions, like how strongly a treatment is needed or how it should be distributed through society.

Experiments also can't help when we start from an effect and search backward for its cause — as with a new disease of unknown origin. That kind of detective work, likened to how detectives match a crime's modus operandi to a suspect, or how epidemiologists run a case-control study, is a different kind of science than the forward-looking logic of an experiment.

Causal Description Versus Causal Explanation

Experimentation's unique strength is causal description — describing the consequences attributable to deliberately varying a treatment. It does less well at causal explanation — clarifying the mechanisms and conditions under which that relationship holds.

Most children quickly learn that flicking a switch turns on a light — a causal description. Almost none could explain why, by decomposing the switch and the light into their essential, causally active parts and tracing the mediating process (the flow of electricity, the excitation of photons) that connects them.

Why Explanation Matters

Causal explanation becomes urgent exactly when a simple description fails — when the light switch stops working and flicking it harder doesn't help. Explanatory knowledge tells us which features of a cause-and-effect relationship are essential and which can be swapped out.

Descriptive causation usually concerns simple relationships between molar (whole-package) treatments and outcomes — that psychotherapy decreases depression, say. Explanatory causation breaks those molar packages into their molecular parts, discovering, for instance, that the verbal interactions and placebo features of therapy reduce depression's cognitive symptoms, while payment for services does not.

Explanation in Practice

Description and explanation aren't as cleanly separated in practice as in the abstract. Many causal explanations are really chains of descriptive links, and experiments can test each link, compare competing theories, or examine moderators and mediators.

Explanation isn't always required for practical impact, though. Economist Richard Lewontin argued that knowing the DNA sequence of a defective gene doesn't by itself let us intervene — intervention requires the normal protein in the right place, cells, and amount. Practical solutions to real problems can take decades of applied follow-up work after a mechanism is understood, or even without full explanation at all.

Balancing Description and Explanation

In the end, causal description and causal explanation exist in delicate balance. Experiments do best at improving descriptions; they do less well at explaining why those descriptions hold.

Still, many causal descriptions are dependable and strong enough on their own to be worth building policy and theory around. That school desegregation causes white flight, that outgroup threat causes ingroup cohesion, that psychotherapy improves mental health, or that diet reduces the retardation caused by PKU: such dependable relationships are useful to policymakers, practitioners, and scientists alike, even without complete explanations of why they hold.

VOCABULARY, 1 OF 9

The Vocabulary of Experiments

Some terms are precise and consistently used; others are blurred. But one attribute is common to all experiments: control over the application of treatment.

Experiment

A study in which an intervention is deliberately introduced to observe its effects.

Randomized Experiment

Units are assigned to treatment or comparison by a random process, like a coin toss.

Quasi-Experiment

An experiment in which units are not assigned to conditions randomly.

Natural Experiment

Not really an experiment, since the cause usually can't be manipulated; contrasts a naturally occurring event with a comparison condition.

Correlational Study

Usually synonymous with nonexperimental or observational study; simply observes the size and direction of a relationship.

The Randomized Experiment

The most clearly described variant is the randomized experiment, widely credited to Ronald Fisher, first used in agriculture and later spread to other fields because it promised control over extraneous variation without needing a physical laboratory.

Its distinguishing feature: the treatments being compared, including no treatment at all, are assigned to units by chance. Done correctly, random assignment creates groups that are probabilistically similar to each other on average — so any outcome difference observed at the end is likely due to treatment, not pre-existing differences. This is why it is often called the gold standard for treatment outcome research.

The Quasi-Experiment

Quasi-experiments share the same purpose as randomized experiments — testing causal hypotheses about manipulable causes — but by definition lack random assignment.

Assignment instead happens through self-selection (people choosing their own treatment) or administrator selection (a teacher, physician, or legislator deciding who gets what). Researchers using quasi-experiments can still control a great deal — how measures are scheduled, which comparison groups are used, how nonrandom assignment is carried out — even without full control over exposure itself.

Ruling Out Alternative Explanations

Because quasi-experimental groups can differ systematically in ways other than treatment, researchers have to enumerate alternative explanations for an observed effect one by one, then use logic, design, and measurement to rule each one out.

Suppose a volunteer group of children gets a new reading program and a group of nonvolunteers doesn't. If the treatment group does better, is it because of the program, or because volunteers were already developing faster? Or because they had more books at home? Multiple pretests, or directly measuring home reading resources, can help assess these alternatives — but the more plausible alternatives there are, the more demanding the quasi-experiment becomes.

The Logic of Falsification

Ruling out alternative explanations is closely related to the falsificationist logic popularized by philosopher Karl Popper. It's logically difficult to ever be sure a general conclusion, like “all swans are white,” is correct — but a single black swan is enough to disprove it.

Popper urged scientists to try to falsify the conclusions they favor, rather than only seeking support for them. Quasi-experimentation works the same way: it requires researchers to identify a causal claim and then actively generate and examine plausible alternative explanations that might falsify it.

The Limits of Falsification

Philosopher Thomas Kuhn pointed out that falsification depends on two assumptions that can never be fully met: that a causal claim is perfectly specified, and that its measures are perfectly valid reflections of the underlying theory.

Since all observation is theory-laden, an apparent disconfirmation can often be explained away by respecifying the theory instead. A fallibilist version of falsification survives this critique: causal studies can still usefully improve our understanding of general trends, even without ever achieving fully definitive, theory-free confirmation or disconfirmation. A finding becomes more factlike when it keeps recurring across different theories, measurements, settings, and times.

Plausible Versus Possible Alternatives

It's neither feasible nor desirable to rule out every possible alternative explanation for a causal relationship — only the plausible ones, since the number of possible alternatives is endless.

But plausibility itself can be deceiving. Stomach ulcers were long attributed to stress and excess acid, since scientists assumed an acid-filled stomach would destroy any pathogen. In 1982, researchers discovered spiral bacteria, later named *H. pylori*, in ulcer patients' stomachs — and by 1994, a US health consensus panel concluded *H. pylori* was the major cause of most peptic ulcers. What counts as plausible depends on social consensus and evidence, not just logical possibility, and it can change.

The Natural Experiment

A natural experiment describes a naturally occurring contrast between a treatment and a comparison condition, often one that isn't even potentially manipulable — like researchers examining whether California earthquakes caused drops in property values.

Economists once had great faith in statistical adjustment to make nonexperimental comparisons valid, but studies of job training programs showed those adjustments produced unstable estimates that didn't match randomized results. So economists turned to natural experiments instead — such as studying the Miami job market after a wave of prisoners was suddenly released from Cuban jails — assuming the timing of the release was independent of the economic trends it was compared against.

Nonexperimental Designs

Correlational, passive observational, and nonexperimental design all describe situations where a presumed cause and effect are identified and measured, but structural features of experiments — random assignment, pretests, control groups — are missing.

Instead, researchers rely on measuring alternative explanations individually and statistically controlling for them. For a nonexperimental causal claim to be credible, the important alternative explanations must be known, measured well, and handled with an appropriately specified model. In cross-sectional studies, where all data are gathered at one time, researchers may not even know whether the cause preceded the effect. These are difficult conditions to meet well, which is why many commentators doubt that nonexperimental designs can support strong causal inferences in most cases.

The Generalization Problem

The strength of experimentation is its ability to illuminate causal inference within a single study. Its weakness is doubt about how far that causal relationship generalizes beyond the study that found it.

This is a defining focus of the broader theory this book develops. Here, the chapter introduces the core generalization issues that later chapters expand on in depth.

GENERALIZATION, 2 OF 8

Every Experiment Is a Local Event

Most experiments are highly localized and particularistic: conducted in a restricted range of settings, often just one; using a particular version of a treatment rather than a full sample of possible versions; using a convenient sample of people rather than a well-described population; and fixed at one point in time.

Yet readers of an experiment are rarely interested in that one particular, past, local study. They usually want to learn about a theoretical construct or a broader policy — which requires generalizing well beyond the specific operations used in the original study. But broad generalization is not automatically the goal: a narrowly occurring effect can still be scientifically important if it demonstrates that something is possible or helps test a theory.

Two Kinds of Generalization

Researcher Lee Cronbach described every experiment as made up of units, treatments, observations, and settings — and identified two distinct generalization problems.

Construct validity asks: what are the higher-order constructs that the actual units, treatments, observations, and settings used in a study represent? External validity asks: does the causal relationship hold over variation in persons, settings, treatments, and outcomes not directly studied? This chapter's theory of causal generalization builds on Cronbach's framework, combined with the authors' own earlier work.

GENERALIZATION, 4 OF 8

Construct Validity: Generalization as Representation

The first generalization problem is how to go from the particular units, treatments, observations, and settings sampled in an experiment to the higher-order constructs they're meant to represent.

In a study by Fortin and Kirouac, nurses gave surgical patients a hospital tour and basic facts about their upcoming operation, then measured recovery using an activities-of-daily-living scale and painkiller use. The real question is whether these particular details actually represent the target constructs researchers care about — here, whether patient education (the cause) promotes physical recovery (the effect) among surgical patients generally (the population) in hospitals generally (the setting).

When Reconceptualization Happens

Empirical results often force researchers to revise their initial understanding of what a study actually represents — sometimes narrowing the claim, sometimes broadening it.

In the patient-education example, if the informational content of the hospital tour, but not the tour itself, turns out to drive recovery, the cause might be narrowed to informational patient education. Conversely, later readers might reinterpret the same study more broadly — treating patient education as a subclass of interventions that increase perceived control, or perceived control as a special case of the still more general construct of self-efficacy.

GENERALIZATION, 6 OF 8

External Validity: Generalization as Extrapolation

The second generalization problem is whether a causal relationship holds over variations in persons, settings, treatments, and outcomes. Someone reading about a Head Start program's effect on reading scores among poor Black children in 1980s Memphis might wonder if a similar program would improve math scores for poor Hispanic children in Dallas today.

This reminds us that generalization is not a synonym for broader application. There's no presumption that Dallas is somehow a broader population than Memphis, or Hispanic children a broader category than Black children — generalization can run from one kind of setting or clientele to another very different one, not just from narrow to broad.

GENERALIZATION, 7 OF 8

Narrow to Broad, and Broad to Narrow

Some generalizations do run from narrow to broad — a researcher who randomly samples from a national population can generalize probabilistically to the rest of that population, which is exactly the rationale for random selection.

But generalization can also run from broad to narrow. Cronbach described a study comparing private- and public-school student performance: individual parents care which type of school is better for their particular child, not for students in general. Whether narrow to broad, broad to narrow, or across units at similar levels, all these questions share the same underlying need — inferring how far an effect holds across variation.

The Limits of Sampling

The method most often recommended for close-fitting generalizations is formal probability sampling — but it presupposes clearly defined populations of units, treatments, observations, and settings to sample from, which is rarely available.

Random selection of people is uncommon enough in cause-probing research; random selection of treatments, outcomes, and settings is rarer still, since no definitive list of possible treatments usually exists. Two more informal approaches are often more practical: purposive sampling of heterogeneous instances (deliberately including diverse cases) and purposive sampling of typical instances (selecting cases that resemble the class mode) — though neither is backed by the statistical logic of formal random sampling.

Random Selection

Chooses cases to represent a population — helps generalization.

Random Assignment

Places cases into treatment conditions — helps causal inference.

GROUNDING THEORY, 1 OF 3

A Grounded Theory of Causal Generalization

Sampling methods alone can't solve either generalization problem — formal probability sampling requires populations that are rarely defined, and purposive sampling isn't backed by a statistical logic that justifies generalization.

Practicing scientists routinely make causal generalizations without formal probability sampling. So this chapter presents a theory of generalization grounded in how scientists actually practice science, built around five closely related principles.

GROUNDING THEORY, 2 OF 3

The Five Principles of Generalization

Surface Similarity

Assessing the apparent similarities between a study's operations and the prototypical characteristics of the target.

Ruling Out Irrelevancies

Identifying which features of a study are irrelevant because they don't change a generalization.

Making Discriminations

Clarifying the key distinctions that limit how far a generalization extends.

Interpolation and Extrapolation

Interpolating to unsampled values within the studied range, and the harder task of extrapolating beyond it.

Causal Explanation

Developing and testing explanatory theories about the mediating processes essential to transferring a causal relationship.

GROUNDING THEORY, 3 OF 3

How Scientists Actually Generalize

Scientists use these five principles constantly, and almost automatically, throughout every phase of research — not just with single studies, but across entire literatures.

When reading a published study, scientists ask whether it resembles their own planned work closely enough to be relevant. When designing a new study, they anticipate whether their planned sample matches the prototypical features of the constructs they're curious about. During analysis, they test whether their construct descriptions still fit the data. And in literature reviews, they apply all five principles to judge what kinds of generalizations a whole body of research can actually support.

Experiments and Metascience

Extensive philosophical debate surrounds experimentation. In one sense, this debate is incidental to the daily practice of science — experimentation is as old as humanity, and its place in science is probably assured forever.

Still, understanding these debates helps scientists see the real limits of experimentation more clearly. This section reviews some of the most influential critiques, from the history, philosophy, and sociology of science, and what they mean in practice.

The Kuhnian Critique

Philosopher Thomas Kuhn described scientific revolutions as abrupt shifts between incommensurable paradigms, denying that scientific knowledge accumulates gradually and undermining any claim that experiments provide firm causal knowledge.

The Quine-Duhem thesis that grew from this critique holds that the results of any single experiment are inevitably ambiguous — disputable on the grounds that the theoretical assumptions built into the outcome measure, or some other unexamined assumption, might be wrong. Because so many assumptions underlie any scientific test, researchers can always find some to fault, or invent new ones, making theories less testable than their authors intended.

Can Theories Ever Really Be Falsified?

Kuhn's critique is more true of single studies than of entire research programs. But even across a full program of research, undetected biases can still produce flawed conclusions about cause and its generalization.

As a result, no experiment is ever fully certain, and extrascientific beliefs and preferences always have some room to influence the many discretionary judgments built into scientific belief — from how an outcome is measured to which conditions of treatment are studied.

Modern Social-Psychological Critiques

Sociologists working in traditions like social constructivism and epistemological relativism have shown that scientists often fail to follow the norms of “good science” — and that what gets reported as scientific knowledge is partly shaped by social, psychological, and political forces rarely mentioned in published research.

The most extreme version of this view holds that the natural world plays little or no role in constructing scientific knowledge at all. This doesn't deny that real entities exist (ontological realism); it denies that scientific theories reliably describe those entities as they really are (epistemological realism) — a useful counterweight to the naive assumption that studies simply reveal nature directly to us.

Science Runs on Trust

The standard image of a scientist is a skeptic who trusts only personally verified results. That image is only partly right.

Any single study is really an exercise in trust: researchers rely on the vast majority of already-developed methods, findings, and instruments they use without personally verifying them — the ratio of trust to skepticism in a given study is closer to 99-to-1 than the reverse. Historian Steven Shapin has shown that “gentlemanly trust” was central to the establishment of experimental science in 17th-century England, despite its rhetoric of skepticism.

What This Means for Experiments

The net effect of these critiques is a greater appreciation for the equivocal nature of all scientific knowledge. An experiment is not a clear window onto nature; it yields hypothetical, fallible knowledge dependent on context and unstated assumptions.

Most practicing scientists, including this book's authors, describe themselves as ontological realists but weak epistemological relativists — believing both the real world and the influence of ideology, interests, and values shape scientific knowledge. To the extent experiments reveal nature to us, in this view, it is through what one scholar called “a very clouded windowpane.”

The Human Side of Experiments

The experiment is a profoundly human endeavor, affected by all the same human foibles as any other. Scientists notice evidence that confirms their preferred hypotheses more readily than evidence that contradicts them, react to peer and social pressures, and are motivated in part by rewards for their work.

These limits have unique relevance to experimentation specifically: when causal results are ambiguous, as in many weaker quasi-experiments, researchers may attribute causation based on features that have little to do with sound logic or method. Fortunately, these limits aren't insurmountable — formal training can help overcome many of them.

In Defense of Stubborn Facts

These social and psychological critiques have taken some shine off the experiment, and rightly so. But some theorists of science have swung too far, treating experimental evidence as almost irrelevant to theory.

Laboratory scientists place real value in “stubborn facts” with a life span longer than the shifting theories built to explain them — basic results about gravity, for instance, are the same whether framed by Newton's or Einstein's physics. There may be no purely theory-free facts, but some experimental findings are dependable enough, over long enough periods, to be worth treating as if they were. Exploratory experiments not guided by formal theory, and unexpected discoveries tangential to the original hypothesis, have repeatedly produced major scientific advances — a reminder that theory does not determine everything scientists are able to observe.

A World Without Experiments?

Borrowing a thought experiment from philosopher Alasdair MacIntyre: imagine the slate of science and philosophy wiped clean, and we had to reconstruct our understanding of the world from scratch.

Would we reinvent the idea of a manipulable cause? Almost certainly — because of how useful dependable, manipulable causes are for our ability to survive and prosper. Would we reinvent the experiment as a way to study them? Also yes: refining how we conduct experiments and generalize their results is an ongoing process that would eventually rediscover something very much like the experiment itself.

Key Takeaways

A cause is rarely simple. Most causes are inus conditions embedded in a larger, often incompletely understood constellation of contributing factors — and an effect is only knowable by comparison with an unobservable counterfactual.

Experiments excel at causal description but do less well at causal explanation. The key distinctions are whether researchers deliberately control the treatment at all and, when they do, whether assignment to treatment is randomized.

Every experiment is a local event, but research aspires to general conclusions. Generalizing a causal finding — through construct validity and external validity — is not a side issue; it is one of the central, unresolved challenges this book exists to address.

WRAPUP, 2 OF 2

Thank You

An experiment tells us, with unusual confidence, what happened when a cause was deliberately varied in one particular place and time. The harder and more interesting question — and the one this book takes up next — is how far that single finding can be trusted to travel.

Adapted from Chapter 1 of *Experimental and Quasi-Experimental Designs for Generalized Causal Inference* (Shadish, Cook, and Campbell, Houghton Mifflin, 2002) — portfolio project by Omokhodion Alfred Eriakha.